# The Universal Protocol with a level in it: site marking and time-out for spine surgery assisted by a fluoroscopic level map—evidence, and a quality-improvement protocol

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## Abstract

**Background.** The Universal Protocol has been mandatory in accredited United States hospitals since 2004. Wrong-level spine surgery persists under it, and the reason is structural: a time-out verifies a site that has been *marked*, but a vertebral level is not marked, it is counted, on intraoperative images that show a few bodies at a time.

**Evidence.** Two adversarially verified literature syntheses (every claim put to three independent refutation attempts) give the following. Half of responding neurosurgeons report at least one wrong-level operation in their career. Per-procedure incidence is 0.032 % by self-report and 0.12 % in a prospective 34,544-case registry, with the thoracic spine highest at 0.42 %. Chart review found the Universal Protocol would have prevented 62 % of insured wrong-site cases under ideal conditions, and wrong-site events continued at a steady rate through the protocol era. In a contemporary closed-claims series the mean wrong-site indemnity was $136,453 and spine surgery the commonest procedure; 17 % of self-reported wrong-level events led to legal action and 81 % of litigated cases ended in payment. No published study measures the intraoperative localisation *process*—exposures, minutes, dose, needle counts, disagreements—as a quality metric, and no per-event hospital cost exists.

**Proposal.** A single-centre, SQUIRE-structured quality-improvement study of the Universal Protocol performed *with* a level in it. The protocol's three elements map onto a fluoroscopic level map built in theatre: verification enters the target level; site marking is a short sweep of overlapping exposures from the sacrum, linked by software into one map, that places a digital fiducial on the target body in the live image; the time-out confirms the fiducial's level against the surgeon's own count before incision, and a disagreement is resolved by rule. On held-out CT renders the walk from the sacrum places the fiducial on the right body in 94.5 % of 950 walks (96.3 % of those reached) from two exposures on average, and 92.2 % of 7,405 walks on 522 clinical CTs from a second institution; published C-arm repositioning times put the whole manoeuvre at one to three minutes against fifteen for fluoroscopic counting. After a four-month process baseline, the assisted protocol runs for eight months; the primary outcome is the disagreement rate between the surgeon's count and the fiducial, adjudicated against postoperative imaging.

**Conclusion.** The Universal Protocol is necessary and not sufficient for wrong-level spine surgery, because it verifies a mark and the spine has none. Giving it a mark that was named from the

patient's own images, and a time-out that checks that name against the count, is a change to the protocol rather than a replacement for it, and it can be measured at one centre in a year.

## The problem the Universal Protocol was not built for

The Universal Protocol for Preventing Wrong Site, Wrong Procedure, and Wrong Person Surgery requires preprocedure verification, marking of the procedure site, and a time-out; for spinal procedures it allows that intraoperative imaging may be needed to locate and mark the exact vertebral level [The Joint Commission, 2024]. Each element assumes the site can be identified on the patient before incision. For a laterality error that is true: the limb is visible and the mark sits on it. For a vertebral level it is not. The level is identified intraoperatively on fluoroscopy, by placing needles every few levels and counting cephalad from the sacrum on lateral views, or by counting levels on anteroposterior views [Strong et al., 2022, Mayer et al., 2014]. The time-out can confirm that everyone agrees on the target level; it cannot confirm that the count reached it.

Two independent surveys found that about half of responding neurosurgeons had performed at least one wrong-level spine operation [Mody et al., 2008, Groff et al., 2013]. Self-reported incidence was one in 3,110 procedures [Mody et al., 2008]; a prospectively collected multicentre registry of 34,544 degenerative spine operations, in which intraoperative disorientation caused 81 % of the lumbar events, recorded 0.12 % overall, 0.42 % thoracic, 0.13 % lumbar and 0.05 % cervical [Furuya et al., 2026]. A systematic review found twelve incidence studies, none thoracic-specific, and reported that in the one study assessing it, site-verification protocols failed to prevent about one-third of cases [Longo et al., 2012]. The frequently quoted 0.09–4.5 per 10,000 range is wrong-site surgery across all specialties, not a spine rate [Devine et al., 2010].

## What the protocol era shows

Kwaan and colleagues reviewed 13 insured nonspine wrong-site cases from 1985–2004 and judged retrospectively that the Universal Protocol, under optimal conditions, might have prevented eight (62 %) [Kwaan et al., 2006]. Stahel and colleagues analysed 27,370 physician self-reported occurrences in a Colorado insurer database from 2002 to mid-2008 and found 107 wrong-site and 25 wrong-patient procedures, persisting after the Universal Protocol was introduced, with significant harm in 35.5 % of wrong-site cases and one death [Stahel et al., 2010]. The Joint Commission received 127 reports of wrong surgery in 2024, nine wrong-level, seven of those spinal, from a voluntary system that, it cautions, cannot show how often such events actually occur [The Joint Commission, 2025]. NHS England provisionally recorded 185 wrong site surgery Never Events in 2024/25, a category that also counts wrong-site blocks and wrong skin lesions [NHS England, 2025]. None of these series can say how many wrong-level spine operations occur; all of them say the number has not gone to zero.

The costs that are measurable are legal. Seventeen percent of self-reported wrong-level events led to legal action or settlement [Mody et al., 2008]; of 68 litigated wrong-disc-level cases, 55 ended in a payment and surgeons prevailed in 42 % of jury trials [Goodkin and Laska, 2004]; the

mean wrong-site indemnity in a 2013–2020 closed-claims series was $136,453 (median $36,250), with spine and disc surgery the commonest procedure and 45.6 % of claims involving additional surgery [Tan et al., 2023]; malpractice payments for all surgical never events summed to $1.3 billion over 21 years [Mehtsun et al., 2013]; 978 spinal-surgery negligence claims in England over five years were valued at £535.5 million [Machin et al., 2018]. Applying the two per-procedure rates to 1.3–1.6 million spine operations a year gives, as derived arithmetic, roughly 400–1,900 wrong-level operations and $8–37 million in indemnity a year in the United States; the hospital cost of the reoperations, added stay and readmissions those events cause has no published per-event figure, and there is no defensible worldwide estimate.

Table 1: The verified figures and where each comes from. Votes are refutation attempts survived (of 3).

| Figure | Source | Verification |
|---|---|---|
| Half of neurosurgeons report a wrong-level operation | Mody 2008; Groff 2013 | 3/3 twice |
| Incidence 1 in 3,110 (self-report); 0.12 % (registry) | Mody 2008; Furuya 2026 | 3/3 twice |
| Thoracic 0.42 %, lumbar 0.13 %, cervical 0.05 % | Furuya 2026 | 3/3 |
| Universal Protocol would have prevented 62 % | Kwaan 2006 | 2/3 |
| Steady rate through the protocol era | Stahel 2010 | 3/3 |
| 17 % of events litigated; 81 % of litigated paid | Mody 2008; Goodkin 2004 | 3/3, 2/3 |
| Mean wrong-site indemnity $136,453; spine commonest | Tan 2023 | 3/3 |
| $1.3 billion never-event payments, 21 years | Mehtsun 2013 | 3/3 |
| 179 fiducial placements, 51.9 min, 183 mGy | AJNR 2023 | 3/3 |
| Operating-room time $34–37 per minute | Childers 2018 | 3/3 |

## Why counting fails, and what it costs to avoid counting

The traditional intraoperative technique is needles every few levels, counted from the sacrum on lateral fluoroscopy; surgeons disagree even on which structure to count, the facet joint with its pedicle for 67 % and the spinous process for 49 % in the thoracic spine [Strong et al., 2022, Mayer et al., 2014]. The images supply the failure modes. A field of a few bodies means a count to a mid-thoracic level passes through several exposures, each of which can be joined to the previous one body off. A lumbosacral or thoracolumbar transitional vertebra shifts the count for the whole column, and such variants are common enough that any series of a few hundred spines contains them. Ribs are hard to number where a twelfth rib is absent or hypoplastic. Habitus, osteopenia and the scapular shadow degrade the mid-thoracic image most. None of these is a lapse of attention that a time-out addresses.

The cost of avoiding the count is visible at one centre, Mayo Clinic in Rochester, which placed preoperative percutaneous gold fiducial markers in 179 patients over 38 months, 176 of them tho-

racic, at a mean of 51.9 minutes and 183 mGy each, to let the surgeon find the level [Siminski et al., 2023]. At a hospital operating-room cost of $34–37 per minute before physician fees [Childers and Maggard-Gibbons, 2018], each placement represents about $1,750–1,950 of staffed procedural time and the series about 155 hours; the figure is a proxy, since the marker is placed in a fluoroscopy or CT suite, and it omits the second anaesthetic and the second day. That a centre with every resource chooses this over counting is the clearest statement in the literature that the thoracic level is, in practice, un-nameable in theatre.

## The measurement gap

The literature measures outcomes—events, claims, payments—and does not measure the process that produces them. No published study records, per case, how many fluoroscopic exposures were spent on localisation, how many minutes, what dose, how many needles, whether the surgeon and the radiographer disagreed, or how a disagreement was resolved. Without that baseline no intervention can be evaluated, and the one intervention the field has, the time-out, was evaluated retrospectively against claims. A quality-improvement study that measures the process is the missing first step, and it is cheap.

## What the engine does, and what it has been measured at

The level map is built from what the radiographer already does. The first exposure places the sacrum at the bottom of the field; each move is one field minus two vertebral bodies, and the last move is a full move or none. Software links each new frame to the last by image correlation restricted to the step the protocol asked for, so the periodic column cannot lock one body off, and names every body in the map by counting from the sacrum and by a shape reader, with the two readings reconciled. When the target level enters a frame the map marks it: a digital fiducial drawn on the live image, on the body the map says is the target. That is the site mark the Universal Protocol asks for and the spine has never had.

Measured on renders of CT the software was never trained on, walking from the sacrum to every labelled level in turn, the fiducial landed on the right body in 94.5 % of 950 walks on 120 spines (96.3 % of the walks that reached the target) from 2.0 exposures on average, and in 92.2 % of 7,405 walks on 522 clinical CTs from another institution, 96.2 % in the lumbar and 91.6 % in the thoracic spine, from 3.0 exposures. The errors are one body off and are concentrated in the known ambiguities: a spine with four lumbar bodies on the sacrum, or six, where a count from one anchor cannot tell one more lumbar body from one fewer thoracic; the decoder has since gained the four-lumbar alphabet and the remainder are stated as the residual. Accuracy falls with distance from the anchor, 96 % within six levels and about 90 % at ten to sixteen, which is why thoracic work is anchored from C2 or the first rib rather than walked twenty bodies from the sacrum. On 25 real lumbar radiographs, with the detector's own boxes, the map completed on 21; the film domain is where the residual risk is, and it is why the study below keeps the surgeon's count as the decision.

The time is the C-arm's, not the software's: the encoder takes 17 ms per frame and the linking a fraction of a second per walk, while published timed positioning trials put reaching a target view with a C-arm at 36 to 49 seconds on average [Matthews et al., 2007, De Silva et al., 2018]. A lumbar walk of two exposures is therefore about a minute and a mid-thoracic walk of three to four about two to three, against about fifteen minutes for fluoroscopic localization in the thoracic spine and about three with a pre-placed fiducial in one small non-randomised series [Mongardi et al., 2021]. The fiducial moves the work, not removes it: a separate CT procedure of a mean 14.3 minutes 2–48 hours before surgery [Keil et al., 2024], usually as an outpatient the day before [Siminski et al., 2023].

## A quality-improvement protocol (SQUIRE 2.0)

**Setting and population.** One academic neurosurgery service; all elective and urgent thoracic and lumbar operations requiring intraoperative level localisation over twelve months, with cervical cases as a comparison arm. Institutional review as quality improvement.

**Phase 1, baseline (months 1–4).** For every case the circulating radiographer records on a one-page form: the anchor used (sacrum, C2, ribs, preoperative marker), the number of localisation exposures, fluoroscopy minutes and cumulative dose from the C-arm log, the number of needles, whether a preoperative fiducial or CT-guided marker was placed (with its procedure time and dose from the radiology record), the surgeon's stated confidence before incision, and any disagreement between team members and how it was settled. The level reached is adjudicated against postoperative imaging by a radiologist blinded to the form. Transitional anatomy is read from the preoperative CT where one exists.

**Phase 2, the assisted protocol (months 5–12).** The Universal Protocol is performed as written, with each element carried by the map. *Verification:* the target level from the consent and the plan is entered into the system before the patient is draped. *Site marking:* at the start of localisation the radiographer performs the sweep from the sacrum (or from C2 for cervical and upper thoracic work) until the system reports the target in the field, and the digital fiducial is displayed on the live image and on the monitor the team sees. *Time-out:* the surgeon states the level by their own count, the circulator reads the fiducial's level and the system's confidence, and the two are compared aloud with the plan. Agreement proceeds. Disagreement, or a system refusal, invokes a written rule: one more exposure with greater overlap, then a count from the other anchor if one is in reach, then the surgeon's judgement, recorded as such. The surgeon's count remains the decision throughout; the map adds a second, independent reading and a mark. Every disagreement is recorded with its resolution and the adjudicated truth. The system's own accuracy is published separately and is not an outcome here.

**Outcomes.** Primary: the disagreement rate between the surgeon's count and the fiducial at the time-out, and which of the two the adjudicated level supported, by region. Compliance: the share of cases in which all three elements were performed as written, and the share in which the fiducial was in the field at incision. Secondary: localisation exposures, minutes and dose per case, needles per case, preoperative marker placements per month, surgeon confidence, and the

near-miss rate (a count corrected before incision). Balancing: total operative time, and any case in which the map was wrong and the surgeon right.

**Analysis.** Phase 1 gives the process distribution per region, which is the number this literature lacks. Phase 2 is analysed as an interrupted series on exposures and minutes, and the disagreement rate is reported with its interval and every case listed. The study is powered to estimate, not to detect a change in wrong-level events, which at 0.12 % would need tens of thousands of cases; the near-miss rate is the tractable proxy and is reported as such.

Table 2: Process measures recorded per case, and the source of each.

| Measure | Source | Why |
|---|---|---|
| Anchor used | radiographer form | which count was made |
| Localisation exposures, minutes, dose | C-arm log | the cost of counting |
| Needles placed | form | the count's length |
| Preoperative marker (time, dose) | radiology record | the cost of not counting |
| Surgeon confidence before incision | form (5-point) | when the count is doubted |
| Disagreement and resolution | form | the near-miss |
| Level reached | postoperative imaging, blinded | the truth |
| Transitional anatomy | preoperative CT | the known failure mode |

## Limitations

The evidence synthesis is of published figures, several of them self-reported or claims-based, and the annual event and dollar figures are derived from stated rates and volumes rather than counted. The proposed study is single-centre and cannot measure a change in wrong-level events; it measures the process and the near-miss. The second reader is one implementation of a level map, and its own accuracy is a separate question with separate evidence.

## Conclusion

The Universal Protocol verifies a mark. A vertebral level has no mark; it has a count, made on images that show a few bodies at a time, through anatomy that varies. That is why wrong-level spine surgery has persisted through two decades of time-outs, why a centre pays an hour and a second procedure per thoracic patient to avoid counting, and why the next quality-improvement step is not another checklist but a measurement of how the level is named today and a second reader that names it independently.

## References

The Joint Commission. National patient safety goals effective january 2025 for the hospital program. The Joint Commission; Universal Protocol for Preventing Wrong Site,

Wrong Procedure, and Wrong Person Surgery, UP.01.01.01–UP.01.03.01, 2024. URL https://digitalassets.jointcommission.org/api/public/content/9be383450fc941df806b76c5fbdd9ae6

Michael J. Strong, Julianne Santarosa, Timothy P. Sullivan, Noojan Kazemi, Jacob R. Joseph, Osama N. Kashlan, Mark E. Oppenlander, Nicholas J. Szerlip, Paul Park, and Clay M. Elswick. Pre- and intraoperative thoracic spine localization techniques: a systematic review. *Journal of Neurosurgery: Spine*, 36(5):792–799, May 2022. doi: 10.3171/2021.8.SPINE21480. Published online 19 November 2021. PMID: 34798613; PMCID: PMC10193475.

J. E. Mayer, R. P. Dang, G. F. Duarte Prieto, S. K. Cho, S. A. Qureshi, and A. C. Hecht. Analysis of the techniques for thoracic- and lumbar-level localization during posterior spine surgery and the occurrence of wrong-level surgery: results from a national survey. *Spine J*, 14:741–748, 2014. doi: 10.1016/j.spinee.2013.06.068.

M. G. Mody, A. Nourbakhsh, D. L. Stahl, M. Gibbs, M. Alfawareh, and K. J. Garges. The prevalence of wrong level surgery among spine surgeons. *Spine*, 33(2):194–198, 2008. doi: 10.1097/BRS.0b013e31816043d1.

Michael W. Groff, Joshua E. Heller, Eric A. Potts, Praveen V. Mummaneni, Christopher I. Shaffrey, and Justin S. Smith. A survey-based study of wrong-level lumbar spine surgery: the scope of the problem and current practices in place to help avoid these errors. *World Neurosurgery*, 79 (3–4):585–592, 2013. doi: 10.1016/j.wneu.2012.03.017.

Masayuki Furuya, Takahito Fujimori, Shota Takenaka, Yukitaka Nagamoto, Tomiya Matsumoto, Yoshifumi Takahashi, Motoki Iwasaki, and Seiji Okada. Wrong-level spinal surgery: a multicentre analysis of 34,544 consecutive cases. *The Bone & Joint Journal*, 108-B(5):673–680, May 2026. doi: 10.1302/0301-620X.108B5.BJJ-2025-1129.R2. PMID: 42061893.

Umile Giuseppe Longo, Mattia Loppini, Giovanni Romeo, Nicola Maffulli, and Vincenzo Denaro. Errors of level in spinal surgery: an evidence-based systematic review. *The Journal of Bone and Joint Surgery. British volume*, 94-B(11):1546–1550, November 2012. doi: 10.1302/0301-620X.94B11.29553. PMID: 23109637.

Jon Devine, Norman Chutkan, Daniel C. Norvell, and Joseph R. Dettori. Avoiding wrong site surgery: a systematic review. *Spine*, 35(9 Suppl):S28–S36, 2010. doi: 10.1097/BRS.0b013e3181d833ac.

Mary R. Kwaan, David M. Studdert, Michael J. Zinner, and Atul A. Gawande. Incidence, patterns, and prevention of wrong-site surgery. *Archives of Surgery*, 141(4):353–358, April 2006. doi: 10.1001/archsurg.141.4.353. PMID: 16618892. Article 353–357, discussion 357–358. No PMCID.

Philip F. Stahel, Allison L. Sabel, Michael S. Victoroff, Jeffrey Varnell, Alan Lembitz, Dennis J. Boyle, Ted J. Clarke, Wade R. Smith, and Philip S. Mehler. Wrong-site and wrong-patient

procedures in the Universal Protocol era: analysis of a prospective database of physician self-reported occurrences. *Archives of Surgery*, 145(10):978–984, October 2010. doi: 10.1001/archsurg.2010.185. PMID: 20956767.

The Joint Commission. Sentinel event data 2024 annual review. The Joint Commission, Oakbrook Terrace, IL, 2025. URL https://digitalassets.jointcommission.org/api/public/content/eac7511986c0442a9c1ae04b1aa02cc0

NHS England. Provisional publication of Never Events reported as occurring between 1 April 2024 and 31 March 2025. NHS England, Patient safety publication, published 8 May 2025, May 2025. URL https://www.england.nhs.uk/long-read/provisional-publication-of-never-events-reported-as-occu

Robert Goodkin and Lewis L. Laska. Wrong disc space level surgery: medicolegal implications. *Surgical Neurology*, 61(4):323–327, 2004. doi: 10.1016/j.surneu.2003.08.022.

Joy Tan, Jacqueline M. Ross, Daniel Wright, Marc Philip T. Pimentel, and Richard D. Urman. A contemporary analysis of closed claims related to wrong-site surgery. *The Joint Commission Journal on Quality and Patient Safety*, 49(5):265–273, 2023. PMID 36925434.

Winta T. Mehtsun, Andrew M. Ibrahim, Marie Diener-West, Peter J. Pronovost, and Martin A. Makary. Surgical never events in the United States. *Surgery*, 153(4):465–472, 2013. doi: 10.1016/j.surg.2012.10.005.

J. T. Machin, J. Hardman, W. Harrison, T. W. R. Briggs, and M. Hutton. Can spinal surgery in England be saved from litigation: a review of 978 clinical negligence claims against the NHS. *European Spine Journal*, 27(11):2693–2699, 2018. doi: 10.1007/s00586-018-5739-1.

C. P. Siminski, C. M. Carr, D. F. Kallmes, M. P. Oien, J. L. D. Atkinson, J. C. Benson, F. E. Diehn, D. K. Kim, G. B. Liebo, V. T. Lehman, A. A. Madhavan, I. T. Mark, P. P. Morris, D. P. Shlapak, J. T. Verdoorn, and J. M. Morris. Fluoroscopy- and CT-guided gold fiducial marker placement for intraoperative localization during spinal surgery: Review of 179 cases at a single institution—technique and safety profile. *American Journal of Neuroradiology*, 44(5):618–622, 2023. doi: 10.3174/ajnr.A7854. PMID 37080723; PMCID PMC10171395; published online 2023 Apr 20.

Christopher P. Childers and Melinda Maggard-Gibbons. Understanding costs of care in the operating room. *JAMA Surgery*, 153(4):e176233, 2018. doi: 10.1001/jamasurg.2017.6233.

Felix Matthews, Dominik J. Hoigne, Martin Weiser, Guido A. Wanner, Pietro Regazzoni, Norbert Suhm, and Peter Messmer. Navigating the fluoroscope's C-arm back into position: an accurate and practicable solution to cut radiation and optimize intraoperative workflow. *Journal of Orthopaedic Trauma*, 21(10):687–692, 2007. doi: 10.1097/BOT.0b013e318158fd42.

Tharindu De Silva, Joshua Punnoose, Ali Uneri, Mahadevappa Mahesh, Joseph Goerres, Matthew Jacobson, Michael D. Ketcha, Amir Manbachi, Sebastian Vogt, Gerhard Kleinszig, Akhil Jay Khanna, Jean-Paul Wolinsky, Jeffrey H. Siewerdsen, and Greg Osgood. Virtual fluoroscopy for

intraoperative C-arm positioning and radiation dose reduction. *Journal of Medical Imaging*, 5(1): 015005, 2018. doi: 10.1117/1.JMI.5.1.015005.

Lorenzo Mongardi, Jacopo Visani, Giorgio Mantovani, Maria Elena Olivetti, Alba Scerrati, Francesco Cultrera, Luca Ricciardi, Pasquale De Bonis, Michele Alessandro Cavallo, and Giorgio Lofrese. CT guided reference markers for spinal dorsal lesions: A safe and valuable tool impacting intraoperative localization time. *Journal of Clinical Neuroscience*, 84:97–101, February 2021. doi: 10.1016/j.jocn.2020.11.039. PMID 33358493. Epub 2020 Dec 25. No PMCID.

Fee Keil, Frank Hagemes, Matthias Setzer, Bedjan Behmanesh, Gerhard Marquardt, Elke Hattingen, Vincent Prinz, Marcus Czabanka, and Markus Bruder. Minimal invasive pre-op CT-guided gold-fiducials in local anesthesia for easy level localization in thoracic spine surgery. *Journal of Clinical Medicine*, 13(19):5690, 2024. doi: 10.3390/jcm13195690. Article 5690. PMID: 39407750; PMCID: PMC11476588. Published online 2024 Sep 25.